

6 The photograph shows two zebras.



(Source: Caroline Wilcox)

The stripes of zebras are thought to reduce attack by biting flies.

Biting flies cause stress and disease to cattle, reducing food production and causing financial losses for farmers.

(a) Farmers use chemicals to kill the flies.

Flies often evolve resistance to a new chemical within about 10 years.

(i) Explain how flies evolve resistance to new chemicals.

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(ii) Suggest why the evolution of resistance to these chemicals occurs so quickly in flies.

(1)

\*(b) An investigation was carried out to see if the presence of stripes on cattle reduced the number of biting flies on their bodies.

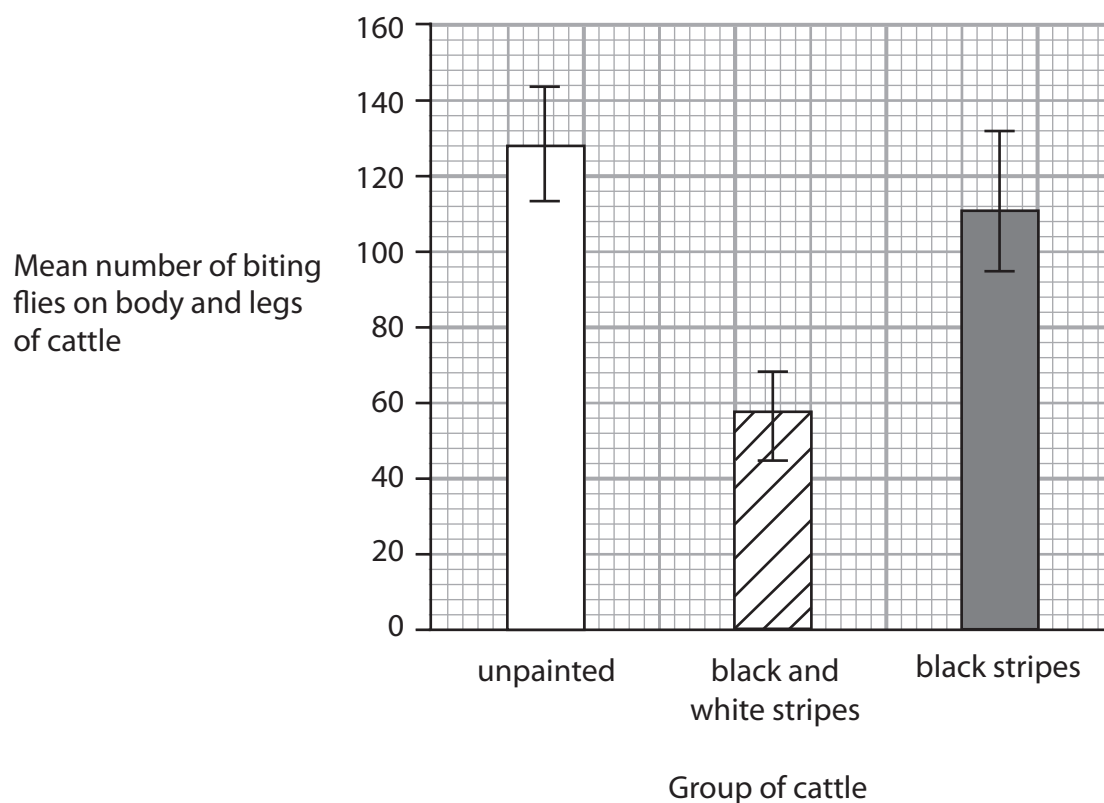
Three groups of black cattle were compared:

- unpainted cattle
- cattle painted with black and white stripes
- cattle painted with black stripes only.

Cattle repel flies by flicking their tails, stamping their feet and skin twitching.

The frequencies of these fly-repelling behaviours were recorded.

The graph and tables show some of the results of this investigation.





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**(Total for Question 6 = 10 marks)**

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